CIVIC INNOVATION & SOFTWARE DEVELOPMENT

CivicTrack

Community Problem Reporting and Resolution Platform

10-Hour Hackathon Project Brief | Problem Statement, Objectives, Workflow & MVP Scope

Project vision
A transparent digital platform through which residents can report local civic problems, the responsible municipal department can acknowledge and resolve them, and overdue complaints are automatically escalated to a Superintendent Officer for intervention.

1. Background

Residents regularly encounter civic infrastructure and public-service problems such as broken streetlights, damaged roads, potholes, overflowing garbage bins, blocked drains, water leakage and unsafe public spaces. These problems may go unreported, be reported through disconnected channels, or remain unresolved without clear communication. Citizens often do not know which department is responsible, whether their complaint has been accepted, who is handling it, or when action will be taken.

Municipal departments also need a structured way to receive complaints, review them, assign responsibility, record progress and identify delays. Without a shared tracking process and escalation mechanism, complaints can be overlooked and public trust can decline.

2. Complete Problem Statement

Develop CivicTrack, a transparent community problem reporting and resolution platform that enables residents to submit local civic complaints with a description, supporting images and an optional location. Each complaint must be routed to the appropriate municipal department for review. When the department accepts the complaint, the system must send an acknowledgement notification to the citizen and make the complaint's progress trackable.

The platform must maintain a clear record of complaint status and departmental actions. If an accepted complaint is not marked as resolved within the defined service period of seven calendar days from the department's acceptance, the system must automatically escalate it to the Superintendent Officer. The Superintendent Officer must be able to review the complaint, see its history and delay, and record the necessary intervention or instructions. The citizen must receive relevant status and escalation updates.

The platform should help administrators categorize and prioritize reports, identify possible duplicate complaints, and view issue distribution through a dashboard or map. Reports must not be treated as verified facts until reviewed by an authorized official, and the system must never claim that an issue is resolved without a recorded status update and supporting action.

3. Project Aim

To build a simple, accountable and trackable digital system that connects citizens with the responsible municipal departments, improves complaint handling, and ensures that unresolved accepted complaints receive supervisory attention through automatic time-based escalation.

4. Project Objectives

- **Accessible reporting:** Allow residents to submit civic complaints using a short description, optional image attachments and optional location.
- **Department routing:** Categorize each complaint and route it to the relevant department for review and acceptance.
- **Citizen acknowledgement:** Send an in-app acknowledgement (and, if configured, email/SMS) when the department accepts a complaint.
- **Progress transparency:** Let citizens view the current status, department response, progress notes and dated history of their complaints.
- **Time-bound accountability:** Start a seven-calendar-day resolution timer when the department accepts the complaint.
- **Automatic escalation:** If the accepted complaint remains unresolved after seven calendar days, automatically move it to the Superintendent Officer's escalation queue and notify the relevant parties.
- **Supervisory action:** Give the Superintendent Officer access to the complaint details, department, elapsed time, status history and controls to record instructions or action.
- **Issue prioritization:** Help administrators identify urgent complaints using category, severity, safety impact and elapsed time.
- **Duplicate awareness:** Identify likely duplicate reports and allow authorized staff to link them while preserving each citizen's original submission.
- **Privacy and auditability:** Protect user information, enforce role-based access and maintain a record of who changed a complaint's status and when.

5. Users and Responsibilities

User / Role	Main responsibilities
Citizen / Resident	Register or sign in; submit a complaint; attach an image; optionally share a location; view acknowledgement and progress; receive escalation updates; provide feedback or request reopening.
Department Officer	View complaints routed to the department; review details; accept or reject with a reason; update progress; add notes; record resolution and supporting evidence.
Superintendent Officer	View automatically escalated overdue complaints; inspect the timeline and department actions; record instructions, intervention, reassignment or other action; monitor closure.
Platform Admin	Manage user roles, department mappings and categories; monitor system activity; maintain configuration and audit records.

6. End-to-End Complaint Workflow

1. Complaint submitted

Citizen enters a description, selects or confirms a category, optionally uploads an image and shares a location. The system creates a unique tracking ID and records the submission time. Status: **Submitted**.

2. Routed for departmental review

The system routes the complaint to the mapped department and places it in that department's pending-review queue. Status: **Pending Department Review**.

3. Department accepts or rejects

The department officer reviews the complaint. On acceptance, the system records the officer and acceptance timestamp, changes the status to **Accepted**, and sends an acknowledgement to the citizen. If rejected, the officer must provide a reason and the citizen is notified.

4. Department works on the issue

The department records assignment and progress notes. Status can move to **In Progress**. Each update is timestamped and visible to the citizen, subject to privacy rules.

5. Seven-day resolution window

The seven-calendar-day timer starts at the department's acceptance timestamp. The system checks whether the complaint has a valid resolved status before the deadline.

6. Automatic escalation

If the complaint is still unresolved at the end of seven calendar days, the system automatically creates an escalation record and places the complaint in the Superintendent Officer's queue. Status: **Escalated**. The citizen and relevant department receive an escalation notification.

7. Superintendent intervention

The Superintendent Officer reviews the complaint, delay and complete action history. The officer records instructions, follow-up, reassignment or other action. The action and timestamp are saved.

8. Resolution and citizen update

The responsible officer records the resolution, adds a note and, where applicable, uploads after-work evidence. The citizen is notified. The citizen can provide feedback or request reopening if the issue remains unresolved.

7. Complaint Status Model

Status	Meaning	Who acts
Submitted	Complaint received and tracking ID generated.	System
Pending Department Review	Waiting for the mapped department to review.	Department Officer
Accepted	Department has accepted responsibility; acknowledgement sent to citizen.	Department Officer / System
In Progress	Work or investigation is underway.	Department Officer
Escalated	Seven-day accepted-complaint period elapsed without resolution; forwarded to Superintendent Officer.	System / Superintendent Officer
Resolved	Authorized officer recorded resolution and update/evidence.	Authorized Officer
Reopened (optional)	Citizen disputes the resolution and requests further review.	Citizen / Department

8. Notification Requirements

Trigger	Recipient	Message / information
Complaint submitted	Citizen	Submission confirmation and unique tracking ID.
Department accepts	Citizen	Complaint accepted by the named department; acceptance date and seven-day target date.
Department rejects	Citizen	Decision and recorded reason; instructions for correction or next steps if applicable.
Progress updated	Citizen	New status and department's progress note.
Seven-day deadline exceeded	Superintendent Officer, department and citizen	Overdue complaint escalation notice, tracking ID, department and elapsed time.
Resolution recorded	Citizen	Resolution update and evidence link/details, where available.

9. Seven-Day Escalation Rules

- The escalation clock starts only when the department accepts the complaint—not merely when the citizen submits it.
- For this MVP, seven days means seven calendar days (24-hour periods), not seven working days. The acceptance timestamp is the source of truth.
- At the deadline, the system checks the latest official status. If the complaint is not Resolved, it is escalated automatically.
- Escalation must be idempotent: repeated scheduler runs must not create multiple escalation records for the same complaint.
- The system must preserve the original department assignment and full timeline after escalation.
- The Superintendent Officer's action must be recorded with officer identity, timestamp and an action note.
- If the department has not accepted or rejected the complaint, it remains pending review. A separate review-time target can be configured later; it is not part of the seven-day resolution timer in this MVP.

Example: Department accepts a complaint on 1 June at 10:00 AM. The seven-calendar-day deadline is 8 June at 10:00 AM. If the complaint is not recorded as Resolved by that time, CivicTrack automatically escalates it to the Superintendent Officer and sends notifications.

10. Core Functional Requirements

ID	Requirement
FR-01	The system shall allow a citizen to submit a complaint with a description and optional image and location.
FR-02	The system shall generate a unique complaint tracking ID and store the submission timestamp.
FR-03	The system shall route a complaint to the appropriate department using its category or configured department mapping.
FR-04	Only an authorized department officer shall accept or reject a complaint assigned to that department.
FR-05	On acceptance, the system shall record the acceptance timestamp and notify the citizen.
FR-06	The system shall show the citizen the current status and chronological complaint timeline.
FR-07	The system shall calculate a seven-calendar-day resolution deadline from the acceptance timestamp.
FR-08	The system shall automatically escalate an accepted complaint that remains unresolved after the seven-day deadline to the Superintendent Officer.
FR-09	The system shall notify the Superintendent Officer, relevant department and citizen when escalation occurs.
FR-10	The Superintendent Officer shall be able to record intervention notes or instructions against an escalated complaint.
FR-11	The system shall record status changes, actor identity and timestamps in an audit/history log.
FR-12	Only authorized officers shall mark a complaint as Resolved; the system shall record a resolution note and optional evidence.

11. Non-Functional Requirements and Constraints

- **Usability:** Complaint submission should be short, mobile-friendly and understandable to non-technical users.
- **Security:** Role-based access must prevent citizens from using department or Superintendent Officer controls.
- **Privacy:** Personal details and precise location must not be exposed publicly without a clear need and user consent.
- **Reliability:** Complaint records and status history must persist in the database; AI or notification failures must not discard submissions.
- **Auditability:** Acceptance, rejection, assignment, progress, escalation and resolution actions must be traceable.
- **Truthful status:** A complaint is not verified merely because it was submitted, and it is not resolved without an authorized recorded update.
- **Time handling:** Store timestamps consistently (preferably UTC) and display them in the configured local timezone.

12. Ten-Hour Hackathon MVP Scope

The goal for a 10-hour hackathon is a working end-to-end demonstration of the complaint lifecycle and automatic escalation—not a complete municipal enterprise platform. Prioritize the core flow before optional AI.

Priority	Feature	10-hour implementation target
Must build	Citizen complaint submission	Description, category, optional image/location, tracking ID; save to database.
Must build	Department review	Department queue with Accept and Reject actions.
Must build	Acceptance notification	In-app notification visible to the citizen after acceptance.

Priority	Feature	10-hour implementation target
Must build	Status tracking	Citizen timeline showing submission, acceptance and progress updates.
Must build	Seven-day escalation	Deadline field plus scheduled/background check; test with a shortened demo interval.
Must build	Superintendent dashboard	Escalated complaint queue and action/instruction entry.
Must build	Resolution update	Authorized officer marks resolved and adds a note; citizen sees notification.
If time remains	AI categorization	Suggest category from complaint text; allow officer correction.
If time remains	Duplicate suggestions / map	Basic same-category, nearby-location match or a simple issue map.
Defer	Advanced vision AI, SMS integration, multilingual voice, predictive analytics	Do not risk the core workflow for these in a 10-hour build.

13. Suggested 10-Hour Execution Plan

Time	Work	Deliverable
Hour 1	Finalize screens, roles, statuses, database fields and team task split.	Agreed scope and project skeleton.
Hours 2–3	Build citizen submission, database persistence and department queue.	Complaint can be submitted and viewed by department.
Hours 4–5	Implement department accept/reject, citizen acknowledgement and status timeline.	Acceptance and progress flow works.
Hours 6–7	Implement seven-day deadline logic, escalation queue and Superintendent Officer action.	Overdue complaint is escalated and can be actioned.
Hour 8	Add resolution workflow, notifications and optional AI categorization if core is stable.	Complete complaint lifecycle.
Hour 9	Test permissions, edge cases, deadline logic and responsive layout.	Critical defects fixed.
Hour 10	Deploy, seed demo data and rehearse the end-to-end presentation.	Working demo and clear pitch.

14. Acceptance Criteria for the Hackathon Demo

- A citizen can submit a complaint and receives a tracking ID.
- The complaint appears in the correct department's review queue.
- The department can accept the complaint, and the citizen sees an acknowledgement notification.
- The citizen can see the complaint's current status and dated updates.
- The system calculates the seven-day deadline from the department acceptance timestamp.
- A test complaint that passes its deadline without resolution appears automatically in the Superintendent Officer's escalation queue.
- The Superintendent Officer can record an action or instruction, and that action appears in the complaint history.
- An authorized officer can record resolution and the citizen can see the update.
- A citizen cannot access department or Superintendent Officer actions.

15. Demo and Testing Note: Simulate Seven Days Safely

Because a hackathon lasts only 10 hours, the team cannot wait seven real days to demonstrate escalation. Implement the actual production rule as seven calendar days, but make the duration configurable for testing. In the demo environment, set the threshold to a short interval (for example, two minutes) or provide a controlled admin-only test action that runs the same escalation-check function. Clearly label this as a demo configuration. Do not hard-code the two-minute interval into the production rule.

The escalation check should be run by a scheduled/background job and should also be callable in tests. It must check: accepted_at is present; status is not Resolved; current time is at or beyond accepted_at + 7 days; and no escalation record already exists. Then create one escalation record, update the status and send notifications.

16. Expected Outcomes

- A centralized channel for residents to report local civic problems.
- Clear visibility into whether the responsible department has accepted a complaint.
- Acknowledgement and progress notifications that keep citizens informed.
- A time-bound escalation mechanism that brings overdue accepted complaints to the Superintendent Officer's attention.
- A traceable record of departmental and supervisory actions.
- A practical prototype that can be extended with maps, AI categorization, duplicate detection and richer analytics.

17. Scope Boundaries and Assumptions

- CivicTrack is a prototype for a hackathon; it is not an official municipal government system unless formally adopted by an authority.
- Department routing is based on a configurable category-to-department mapping. The prototype may use a small set of sample departments.
- The seven-day period is a proposed MVP service target, not a claim about an existing government service-level policy.
- For this project, seven days means calendar days from acceptance. An authority may later configure working days, holidays, pause rules or category-specific service targets.
- The Superintendent Officer is represented as a privileged role in the prototype. Real-world reporting lines and powers must be confirmed with the relevant municipality.
- Notifications in the MVP may be in-app only. SMS, WhatsApp and email integrations are optional extensions.
- AI outputs are suggestions for staff review. They do not establish factual verification or automatically determine final resolution.

Project success definition
The hackathon prototype is successful when a complaint can be submitted by a citizen, reviewed and accepted by the responsible department, acknowledged to the citizen, tracked through progress, automatically escalated after the configured seven-day period if unresolved, acted upon by the Superintendent Officer, and finally updated with a recorded resolution.

End of Project Brief